

COMP 2322 Computer Networking

Multi-thread Web Server

Project Report

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1 Overview

This project implements a multi-threaded web server from scratch using Python socket programming. Server runs on port **8848**.

1.1 Project Structure

```
project/
report/
  report.tex
  Screenshots/
    01_server_startup.png
    02_stage1_request.png
    03_get_text.png
    04a_get_image_browser.png
    04b_get_image_curl.png
    05_head_method.png
    06_404_not_found.png
    07_403_forbidden.png
    08_400_bad_request.png
    09_10_lastmod_304.png
    11_12_keepalive_close.png
    13_concurrent_load.png
    14_access_log.png
    15_status_dashboard.png
  server.py
  webs/
    index.html
    test.jpg
  logs/
    server.log
  test.ipynb
```

2 Design Highlights

2.1 Producer-Consumer Thread Model

Main thread acts as dispatcher, only accepting connections. Worker threads handle requests independently. This prevents slow requests from blocking new connections.

2.2 Short-Circuit Cache Validation

Server checks If-Modified-Since before reading files. If unchanged, returns 304 immediately, skipping disk I/O and network transmission.

2.3 Two-Layer Path Traversal Protection

1. `os.path.normpath()` normalizes malicious paths

2. `os.path.realpath()` prefix verification ensures file stays within web root

2.4 Structured Logging

Pipe-separated format enables easy parsing: `ip | timestamp | request | status`

2.5 Zero-Config Startup

Server auto-creates `webs/`, `logs/`, and default `index.html` on first run.

3 Implementation - Three Stages

3.1 Stage 1: Basic Socket Server

Simple TCP server that accepts connections and returns a fixed response.

Listing 1: Basic Server Loop

```
1 def main():
2     server_socket = socket.socket(socket.AF_INET, socket.
        SOCK_STREAM)
3     server_socket.bind(('127.0.0.1', 8848))
4     server_socket.listen(5)
5     while True:
6         client_socket, addr = server_socket.accept()
7         data = client_socket.recv(4096)
8         client_socket.send(b"HTTP/1.1 200 OK\r\n\r\nHello")
9         client_socket.close()
```

3.2 Stage 2: File Serving

Added file reading, MIME types, and status codes (200, 404, 403).

Listing 2: File Serving

```
1 file_path = os.path.join('webs', path.lstrip('/'))
2 if os.path.exists(file_path):
3     with open(file_path, 'rb') as f:
4         content = f.read()
5     send_response(200, content)
6 else:
7     send_response(404, "Not Found")
```

3.3 Stage 3: Multi-threading & Advanced Features

- Each request handled in separate thread
- HEAD method support
- Logging to `logs/server.log`

Listing 3: Multi-threading

```
1 threading.Thread(target=handle_client, args=(client_socket, addr))
   .start()
```

4 Additional Features

4.1 Cache Validation (304)

Server returns Last-Modified header. If client sends If-Modified-Since and file unchanged, returns 304.

4.2 Keep-Alive

Same TCP connection reused for multiple requests. Loop continues until Connection: close.

4.3 Status Dashboard

Visit <http://127.0.0.1:8848/status> to see:

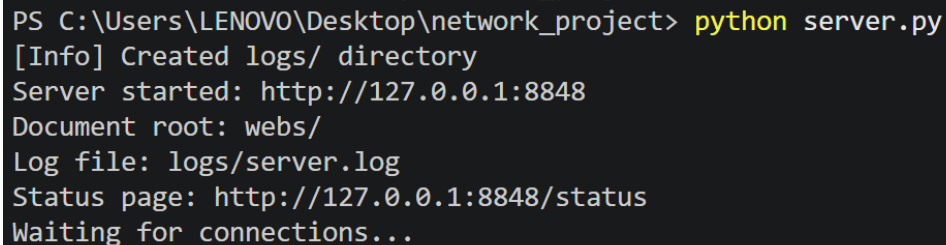
- Server uptime
- Active threads
- Total requests
- Recent logs

5 Test Results

5.1 Test 1: Server Startup

Command: `python server.py`

Expected: Server starts on port 8848, creates webs/ and logs/ directories.



```
PS C:\Users\LENOVO\Desktop\network_project> python server.py
[Info] Created logs/ directory
Server started: http://127.0.0.1:8848
Document root: webs/
Log file: logs/server.log
Status page: http://127.0.0.1:8848/status
Waiting for connections...
```

Figure 1: Server startup

5.2 Test 2: Stage 1 Request Display

Command: Client connects, server displays raw HTTP request.

Expected: Server console shows received HTTP request.

```
[New connection] From: ('127.0.0.1', 61496)
[Active threads] 1
[Request] GET /
[New connection] From: ('127.0.0.1', 55385)
[Active threads] 2
```

Figure 2: Stage 1 - Raw request display

5.3 Test 3: GET Text File

Command: `curl http://127.0.0.1:8848/index.html`

Expected: HTTP 200 OK, returns index.html content.

```
C:\Users\LENOVO>curl http://127.0.0.1:8848/index.html
<html>
<body>
  <h1>There will be BIRDS...</h1>
  <p>Gu~Gu~Gu~</p>
</body>
</html>
```

Figure 3: GET text file - 200 OK

5.4 Test 4: GET Image File

Command: `curl -I http://127.0.0.1:8848/test.jpg`

Expected: HTTP 200 OK, Content-Type: image/jpeg. Browser displays image.

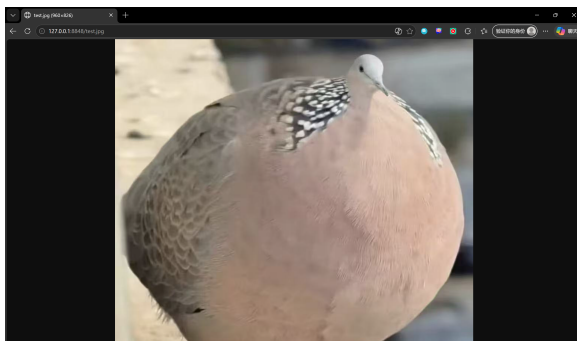


Figure 4: Browser displaying image (Spotted Dove)

```
C:\Users\LENOVO>curl -I http://127.0.0.1:8848/test.jpg
HTTP/1.1 200 OK
Content-Type: image/jpeg
Content-Length: 46161
Last-Modified: Mon, 20 Apr 2026 12:38:06 GMT
Connection: keep-alive
```

Figure 5: curl -I: 200 OK, image/jpeg

5.5 Test 5: HEAD Method

Command: `curl -I http://127.0.0.1:8848/index.html`

Expected: HTTP 200 OK with headers only, no message body.

```
C:\Users\LENOVO>curl -I http://127.0.0.1:8848/index.html
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 87
Last-Modified: Mon, 20 Apr 2026 14:32:05 GMT
Connection: keep-alive
```

Figure 6: HEAD method - headers only

5.6 Test 6: 404 Not Found

Command: `curl http://127.0.0.1:8848/nonexistent.html`

Expected: HTTP 404 Not Found with error page.

```
C:\Users\LENOVO>curl http://127.0.0.1:8848/nonexistent.html

<html>
<head><title>404 Not Found</title></head>
<body>
  <h1>404 Not Found</h1>
  <p>File '/nonexistent.html' not found</p>
  <hr>
  <p>Multi-Thread Web Server | <a href="/status">Server Status</a></p>
</body>
</html>
```

Figure 7: 404 Not Found

5.7 Test 7: 403 Forbidden

Command: `curl --path-as-is http://127.0.0.1:8848/../server.py`

Expected: HTTP 403 Forbidden (path traversal protection).

```
C:\Users\LENOVO>curl --path-as-is http://127.0.0.1:8848/../server.py

<html>
<head><title>403 Forbidden</title></head>
<body>
  <h1>403 Forbidden</h1>
  <p>Access denied</p>
  <hr>
  <p>Multi-Thread Web Server | <a href="/status">Server Status</a></p>
</body>
</html>
```

Figure 8: 403 Forbidden - path traversal protection

5.8 Test 8: 400 Bad Request

Command: Python socket (see Jupyter Notebook Cell 7)

Expected: HTTP 400 Bad Request.

Cell 7

Purpose: Send an invalid raw socket request and verify the `400 Bad Request` response.

```
print("--- Test Item: 400 Bad Request ---")
try:
    s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    s.connect(("127.0.0.1", PORT))
    bad_request = b"HELLO THIS IS NOT HTTP\r\n\r\n"
    s.send(bad_request)
    raw_response = s.recv(1024).decode("utf-8", errors="ignore")
    s.close()

    status_line = raw_response.split("\r\n")[0]
    print("Sent invalid request: HELLO THIS IS NOT HTTP")
    print(f"Server response status line: {status_line}")
except Exception as e:
    print(f"Connection failed: {e}")
```

✓ 0.0s

```
--- Test Item: 400 Bad Request ---
Sent invalid request: HELLO THIS IS NOT HTTP
Server response status line: HTTP/1.1 400 Bad Request
```

Figure 9: 400 Bad Request (Cell 7 output)

5.9 Test 9-10: Last-Modified and 304 Not Modified

Step 1: Get Last-Modified header

```
curl -v http://127.0.0.1:8848/index.html 2>&1 | findstr "Last-Modified"
< Last-Modified: Mon, 20 Apr 2026 14:50:23 GMT
```

Step 2: Conditional request with If-Modified-Since

```
curl -v --header "If-Modified-Since: Mon, 20 Apr 2026 14:50:23 GMT" http://127.0.0.1:8848/index.html
< HTTP/1.1 304 Not Modified
< Connection: close
```

```
C:\Users\LENOVO>curl -v http://127.0.0.1:8848/index.html 2>&1 | findstr "Last-Modified"
< Last-Modified: Mon, 20 Apr 2026 14:50:23 GMT

C:\Users\LENOVO>curl -v --header "If-Modified-Since: Mon, 20 Apr 2026 14:50:23 GMT" http://127.0.0.1:8848/index.html
* Trying 127.0.0.1:8848...
* Established connection to 127.0.0.1 (127.0.0.1 port 8848) from 127.0.0.1 port 61558
* using HTTP/1.x
> GET /index.html HTTP/1.1
> Host: 127.0.0.1:8848
> User-Agent: curl/8.18.0
> Accept: */*
> If-Modified-Since: Mon, 20 Apr 2026 14:50:23 GMT
>
* Request completely sent off
< HTTP/1.1 304 Not Modified
< Connection: close
* shutting down connection #0
```

Figure 10: Last-Modified header and 304 Not Modified response

5.10 Test 11-12: Connection Keep-Alive and Close

Source: Jupyter Notebook Cell 9

Test A: Keep-Alive - Connection header handling

Sent header: {'Connection': 'keep-alive'}

Returned Connection header: keep-alive

Test B: Connection Close

Sent header: {'Connection': 'close'}

Returned Connection header: close

Cell 9

Purpose: Test how the server handles the `Connection` header for both `keep-alive` and `close`.

```
print("--- Test Item: Connection header ---")
url = f"{BASE_URL}/index.html"

headers_keep = {"Connection": "keep-alive"}
r_keep = requests.get(url, headers=headers_keep)
print(f"Sent header: {headers_keep}")
print(f"Returned Connection header: {r_keep.headers.get('Connection')}")
print()

headers_close = {"Connection": "close"}
r_close = requests.get(url, headers=headers_close)
print(f"Sent header: {headers_close}")
print(f"Returned Connection header: {r_close.headers.get('Connection')}")

--- Test Item: Connection header ---
Sent header: {'Connection': 'keep-alive'}
Returned Connection header: keep-alive

Sent header: {'Connection': 'close'}
Returned Connection header: close
```

Figure 11: Connection header handling - keep-alive and close (Cell 9)

5.11 Test 13: Multi-threading Concurrent Load

Command: 10 simultaneous GET requests via Python threading.

Expected: All 10 requests return 200 OK.

```
C:\Users\LENOVO>for /l %i in (1,1,10) do start /b curl http://127.0.0.1:8848/index.html
C:\Users\LENOVO>start /b curl http://127.0.0.1:8848/index.html
C:\Users\LENOVO>start /b curl http://127.0.0.1:8848/index.html
C:\Users\LENOVO>start /b curl http://127.0.0.1:8848/index.html
<html>
<body>
  <h1>There will be BIRDS...</h1>
  <p>Gu~Gu~Gu~</p>
</body>
</html>
C:\Users\LENOVO>start /b curl http://127.0.0.1:8848/index.html
<html>
<body>
  <h1>There will be BIRDS...</h1>
  <p>Gu~Gu~Gu~</p>
</body>
</html>
C:\Users\LENOVO>start /b curl http://127.0.0.1:8848/index.html
<html>
<body>
  <h1>There will be BIRDS...</h1>
  <p>Gu~Gu~Gu~</p>
</body>
</html>
C:\Users\LENOVO>start /b curl http://127.0.0.1:8848/index.html
<html>
<body>
  <h1>There will be BIRDS...</h1>
  <p>Gu~Gu~Gu~</p>
</body>
</html>
C:\Users\LENOVO>start /b curl http://127.0.0.1:8848/index.html
<html>
<body>
  <h1>There will be BIRDS...</h1>
```

Figure 12: 10 concurrent requests - all successful

5.12 Test 14: Access Log

Command: type logs\server.log (Windows) or cat logs/server.log

Expected: Each request recorded with IP, timestamp, request, status code.

```

C:\Users\LENOVO\Desktop\network_project>type logs\server.log
127.0.0.1 | 2026-04-10 23:23:42 | GET /index.html | 200
127.0.0.1 | 2026-04-14 22:36:11 | GET /index.html | 200
127.0.0.1 | 2026-04-14 22:36:11 | GET /favicon.ico | 404
127.0.0.1 | 2026-04-14 22:36:45 | HEAD /index.html | 200
127.0.0.1 | 2026-04-14 22:36:55 | HEAD /index.html | 200
127.0.0.1 | 2026-04-20 20:20:51 | GET /status | 200
127.0.0.1 | 2026-04-20 20:21:22 | GET /status | 200
127.0.0.1 | 2026-04-20 20:21:22 | GET /favicon.ico | 404
127.0.0.1 | 2026-04-20 20:21:29 | GET /status | 200
127.0.0.1 | 2026-04-20 20:21:34 | GET /status | 200
127.0.0.1 | 2026-04-20 20:38:55 | GET /test.jpg | 200
127.0.0.1 | 2026-04-20 20:38:55 | GET /favicon.ico | 404
127.0.0.1 | 2026-04-20 21:07:21 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:07:33 | GET /test.png | 404
127.0.0.1 | 2026-04-20 21:07:56 | GET /test.jpg | 200
127.0.0.1 | 2026-04-20 21:08:04 | HEAD /index.html | 200
127.0.0.1 | 2026-04-20 21:08:17 | GET /this_file_does_not_exist.html | 404
127.0.0.1 | 2026-04-20 21:08:26 | GET ../server.py | 403
127.0.0.1 | 2026-04-20 21:08:39 | HELLO THIS | 404
127.0.0.1 | 2026-04-20 21:08:54 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:08:54 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:00 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:00 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:03 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:03 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:03 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:03 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:03 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:03 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:46 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:09:53 | GET /test.jpg | 200
127.0.0.1 | 2026-04-20 21:20:39 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:20:39 | GET /test.jpg | 200
127.0.0.1 | 2026-04-20 21:20:39 | HEAD /index.html | 200
127.0.0.1 | 2026-04-20 21:20:39 | GET /this_file_does_not_exist.html | 404
127.0.0.1 | 2026-04-20 21:20:39 | GET ../server.py | 403
127.0.0.1 | 2026-04-20 21:20:39 | HELLO THIS | 404
127.0.0.1 | 2026-04-20 21:20:39 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:20:39 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:20:39 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:20:39 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:20:39 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:20:39 | GET /index.html | 200
127.0.0.1 | 2026-04-20 21:20:39 | GET /index.html | 200

```

Figure 13: Access log file contents

5.13 Test 15: Status Dashboard

Method: Visit <http://127.0.0.1:8848/status> in Browser

Expected: HTML dashboard showing uptime, active threads, request count, recent logs.

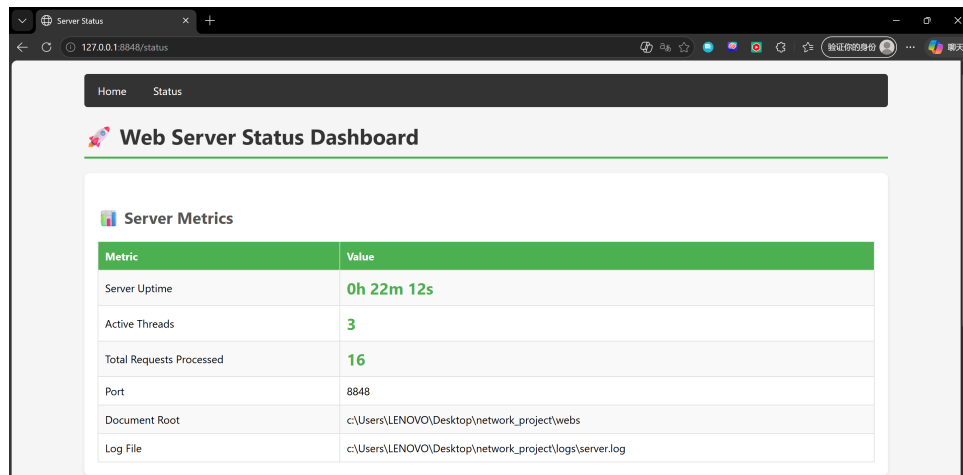


Figure 14: Status dashboard (browser view)

6 Conclusion

Successfully implemented a fully functional multi-threaded web server supporting HTTP/1.1 features including persistent connections, cache validation, and concurrent request handling.